



Assessment Template

Computing

Assessment:	2
Assignment Task	2
Overall marks will be awarded according to the following criteria:	2
Resources	2
Grading Criteria	3
General Requirements for Students:	4

Assessment:

Item	Description
Module Title:	Graph & AI
Module Code:	B9AI101
Module Lecturer (s):	Terri Hoare
Programme/Cohort:	MSc Artificial Intelligence
Method of Assessment:	Oral
Percentage (%) Weighting:	50%
MIMLOs being assessed:	MIMLO01; MIMLO02; MIMLO04; MIMLO05
Assessment Number:	2
Individual/Group:	Group Artefact and Presentation; Individual Report
Issue Date:	Reading Week
Submission Date:	End Week 12
Feedback Date:	Week 12 on Presentation
Feedback Strategy:	On Presentation

Assignment Task**Task 1 (Group)**

Working in groups of 2-3 students, select a data source from the Neo4j recommended developer data sources <https://neo4j.com/developer/example-data/> and apply the steps of the CRISP-DM data mining methodology to analyze the data.

Business Understanding

Explore the domain use case.

Data Understanding

Explore the Neo4j property graph data model that is used as the knowledge graph for your use case.

Data Preparation

Load the data to Neo4j Sandbox (cloud); Neo4j Community Edition (laptop) or Neo4j Aura (cloud).

****Note** that the real-world data that is available in the sandbox may be used / modified as required for your use case.

Complete an exploratory statistical and visual data analysis of your knowledge graph using Python Py2Neo.

Modelling

Apply appropriate graph algorithms to derive insights from the data. Explore graph embeddings and machine learning use cases on your graph network.

Evaluation

Interpret the insights derived in the context of your problem domain and appropriate metrics for graph learning use cases.

Deployment

Explore how the insights could be deployed (operationalized) in an AI context.

Prepare and deliver a 10-15-minute formal presentation of your case study during the final lecture of the module in Week 12.

(50 marks)

Task 2 (Individual)

Write up your findings for exercise 1 under the CRISP-DM headings.

(50 marks)

Submission

Include a cover page on your report and cite all references. Two files should be loaded to Moodle on or before Sunday of Week 12 (23:55).

1. A SINGLE pdf file named CA02_Surname_First Name_Student-ID
2. A zipped file including your code for task 1.

Overall marks will be awarded according to the following criteria:

001: Task 1 (Group) Selection of Graph Network use case; data modelling; GDS application	25%
002: Task 1 (Group) Statistical and Visual Analysis	15%
003: Task 1 (Group) Use of Tools - Python and Neo4j	10%
004: Task 2 (Individual) Interpretation and insight into algorithm & graph learning use case	50%

Resources

Software or tools required/ useful: Python; Neo4j

Templates, files, data tables provided by the lecturer: Moodle resources

Module Assessment Template

1	2	3	4	5
NO AI	AI-ASSISTED IDEA GENERATION AND STRUCTURING	AI-ASSISTED EDITING	AI TASK COMPLETION, HUMAN EVALUATION	FULL AI
The assessment is completed entirely without AI assistance. This level ensures that students rely solely on their knowledge, understanding, and skills. AI must not be used at any point during the assessment.	AI can be used in the assessment for brainstorming, creating structures, and generating ideas for improving work. No AI content is allowed in the final submission.	AI can be used to make improvements to the clarity or quality of student created work to improve the final output, but no new content can be created using AI. AI can be used, but your original work with no AI content must be provided in an appendix.	AI is used to complete certain elements of the task, with students providing discussion or commentary on the AI-generated content. This level requires critical engagement with AI generated content and evaluating its output. You will use AI to complete specified tasks in your assessment. Any AI created content must be cited.	AI should be used as a 'co-pilot' in order to meet the requirements of the assessment, allowing for a collaborative approach with AI and enhancing creativity. You may use AI throughout your assessment to support your own work and do not have to specify which content is AI generated

		Evaluative Criteria/Performance Level Descriptors (performance?)			
Scale	Grade Band	Criterion 1 - "001"	Criterion 2 - "002"	Criterion 3 - "003"	Criterion 4 - "004"
Exceptional	90-100%	Exceeding module requirements	Exceeding module requirements	Exceeding module requirements	Exceeding module requirements
Outstanding	80-89%	Evidence of non-core advanced knowledge and / or knowledge	Evidence of non-core advanced knowledge and / or interpretation	Evidence of non-core advanced knowledge of tools	Evidence of non-core advanced knowledge / interpretation of findings
Excellent	70-79%	Very good both technically and a well-presented critical interpretation	Very good both technically and a well-presented critical interpretation	Very good both technically and a well-presented	Very good both technically and a well-presented critical interpretation
Very Good	60-69%	Evidence of some critical interpretation beyond that covered in class	Evidence of some critical interpretation beyond that covered in class	Evidence of some knowledge beyond that covered in class	Evidence of some critical interpretation beyond that covered in class
Good	50-59%	Evidence of understanding of core concepts covered in class	Evidence of understanding of core concepts covered in class	Evidence of understanding of core concepts of tool use	Evidence of understanding of core concepts covered in class
Sufficient	40-49%	Submission met basic requirements	Submission met basic requirements	Submission met basic requirements	Submission met basic requirements
Somewhat/ Unfulfilled	30-39%	Submission not completed in full	Submission not completed in full	Submission not completed in full	Submission not completed in full
Unfulfilled	0-29%	Submission not completed in full and poorly presented	Submission not completed in full and poorly presented	Submission not completed in full and poorly presented	Submission not completed in full and poorly presented
MIMLO Alignment (see Module Guide for Minimum Intended Learning Outcomes for Module)					
MIMLO criterion aligned:		MIMLO02	MIMLO04	MIMLO05	MIMLO1,5

General Requirements for Students:

1. All assignments must be submitted no later than the stated deadline (date and time).
2. Assignments submitted after the latest deadline specified (including any approved extension deadline) are considered late and penalised according to the Quality Assurance Handbook (QAH) Part B Section 5.2.2.6 as follows:
 - a. A penalty of 2 marks will be applied per day or part thereof (including weekends and public holidays) for an ongoing failure to submit beyond the submission deadline.
 - b. An examiner has the right to refuse to mark the assignment if the submission instructions have not been observed.
 - c. Where a late assessment is submitted within 14 days of the deadline, and is of a passing standard, the late penalty is capped (such that the minimum grade that can be awarded is 40% for the late submission).
 - d. Where a late assessment is submitted more than 14 days after the deadline, it will receive 0%. The lecturer may, at their discretion, review the submission for feedback.
 - e. Where the assessment is undertaken in a group, the piece of work should be submitted in its complete entirety, and any penalty for late submission incurred applies to all group members.
3. Extensions to assignment submission deadlines will not be granted, other than in exceptional circumstances. To apply for an extension please go to <https://students.dbs.ie/dashboard/SCCM> and open a ticket.
4. All relevant provisions of the Assessment Regulations must be complied with (see QAH B.5).
 - a. Students are required to refer to the assessment regulations in their Programme Handbook, and on the Student Website.
 - b. Dublin Business School penalises students who engage in academic impropriety (i.e. plagiarism, collusion and/or copying, ghost writing/ essay mills, improper use of Generative Artificial Intelligence software).
 - i. Refer to the College's Generative AI Guidelines HERE for further information.
 - c. Guides on referencing are available on the Library website: <https://libguides.dbs.ie/referencing>
 - d. Text-matching analysis software is integrated in Moodle to generate a report regarding the degree of text-matching in a submission.
5. Students are required to retain a copy of each assignment submitted, until the issuing of a transcript indicating the mark awarded and the closure of the Appeal period (2 weeks following the release of final results).
 - a. Results can only be appealed following the release of final results, and the Appeal form must be submitted to the Exams Office within the Appeal period.
 - b. An appeal must be based on valid grounds (see the Appeals Policy QAH B.3.5), dissatisfaction with a grade is not sufficient grounds for an appeal.
 - c. Assignments must be appropriately packaged and presented.
 - d. All assignments should be submitted to your subject/course page on Moodle by the deadline date.

- e. Where a submission involves digital media (i.e formats other than Word, Powerpoint or PDF), it is the submitting students' responsibility to ensure the media is appropriately labelled, fully working and they must retain a copy.
 - f. Components of an assessment which are not included in the final submission cannot normally be subsequently accepted for grading. It is the student's responsibility to ensure their file is uploaded correctly.
 - g. Include an electronic **cover sheet** with the following details to the front of the assignment (see below)
6. Assignments that *breach* the word count requirements will be penalised. *There is a 10% discretion, either way, applicable in terms of word count.*
7. When you submit your assignment you will be asked to click on a button which will declare the following:

By ticking this box I am confirming that this assignment/exam is all my own work. Any sources used have been referenced.

I have read the College rules regarding plagiarism in the QAH Part B Section 3 and understand that penalties will be applied accordingly if work is found not to be my own. All work uploaded is submitted via Ouriginal, whereby a text-matching report will show any similarities with other texts.